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Esercizio 62.

$$\tau_0 = IAB \sin \varphi_0$$

$$\tau_0 = 45 \times 10^{-3} \text{ A} \cdot 24 \times 10^{-4} \text{ m}^2 \cdot 72 \times 10^{-3} \text{ T} \sin(90^\circ) \approx 7.8 \times 10^{-6} \text{ N} \cdot \text{m}$$

$$\tau_1 = IAB \sin(\varphi_1)$$

$$\tau_1 = 45 \times 10^{-3} \text{ A} \cdot 24 \times 10^{-4} \text{ m}^2 \cdot 72 \times 10^{-3} \text{ T} \cdot \sin(30^\circ) \approx 3.9 \times 10^{-6} \text{ N} \cdot \text{m}$$

$$\tau_2 = IAB \sin \varphi_2$$

$$\tau_2 = 45 \times 10^{-3} \text{ A} \cdot 24 \times 10^{-4} \text{ m}^2 \cdot 72 \times 10^{-3} \text{ T} \sin(0^\circ) = 0 \text{ N} \cdot \text{m}$$

Esercizio 65.

$$\mu_m = NIA = 40 \cdot 20 \times 10^{-3} \text{ A} \cdot (0.11 \text{ m})^2 \approx 9.7 \times 10^{-3} \text{ A} \cdot \text{m}^2$$

$$\vec{\tau} = \vec{\mu}_m \times \vec{B} \implies \tau = \mu_m \cdot B = 9.7 \times 10^{-3} \text{ A} \cdot \text{m}^2 \cdot 2.5 \times 10^{-2} \text{ T} \approx 2.4 \times 10^{-4} \text{ N} \cdot \text{m}$$

Esercizio 63.

$$\tau_{\max} = NI_{\max}AB \implies I_{\max} = \frac{\tau_{\max}}{NAB}$$

$$I_{\max} = \frac{5.8 \text{ N} \cdot \text{m}}{1200 \cdot 1.1 \times 10^{-2} \text{ m}^2 \cdot 0.20 \text{ T}} \approx 2.2 \text{ A}$$